Exp. No: 11 Verification of Maxwell's Equations and Displacement Current in Time-Varying Electromagnetic Fields

Aim:

To verify the Ampère–Maxwell law and the point-form Maxwell's equations for time-varying fields by determining the displacement current in a parallel-plate capacitor, comparing conduction and displacement current densities in different media, numerically checking Faraday's law and the Ampère Maxwell law for a uniform plane wave, and computing the magnetic field produced by the displacement current.

Objectives:

1. To determine the displacement current density and displacement current in a capacitor driven by a sinusoidal voltage and show that it equals the conduction current in the leads.

2. To compare the conduction-to-displacement current ratio $\sigma/(\omega\varepsilon)$ for a conductor, a lossy dielectric and a poor conductor and to find the crossover frequency.

3. To verify numerically that the electric and magnetic fields of a uniform plane wave satisfy Maxwell's curl equations.

4. To determine the magnetic field produced by the displacement current between and outside the capacitor plates using the Ampère–Maxwell law.

Test Case (i) – Displacement Current in a Parallel-Plate Capacitor

```
clc; clear;

eps0 = 8.854e-12;

a  = input('Enter plate radius a (m): ');

d  = input('Enter plate separation d (m): ');

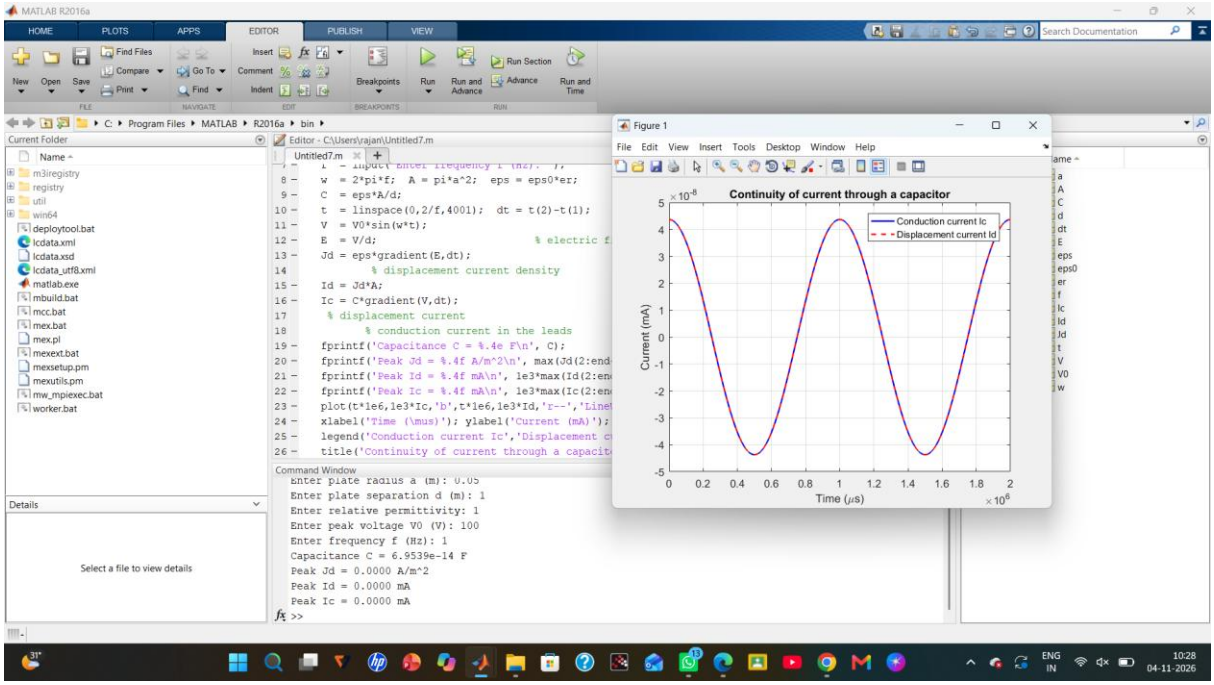
er = input('Enter relative permittivity: ');

V0 = input('Enter peak voltage V0 (V): ');

f  = input('Enter frequency f (Hz): ');

w  = 2*pi*f;  A = pi*a^2;  eps = eps0*er;
```

```
C  = eps*A/d;
t  = linspace(0,2/f,4001);  dt = t(2)-t(1);
V  = V0*sin(w*t);
E  = V/d;                       % electric field between the plates
Jd = eps*gradient(E,dt);
       % displacement current density
Id = Jd*A;
Ic = C*gradient(V,dt);
 % displacement current
      % conduction current in the leads
fprintf('Capacitance C = %.4e F\n', C);
fprintf('Peak Jd = %.4f A/m^2\n', max(Jd(2:end-1)));
fprintf('Peak Id = %.4f mA\n', 1e3*max(Id(2:end-1)));
fprintf('Peak Ic = %.4f mA\n', 1e3*max(Ic(2:end-1)));
plot(t*1e6,1e3*Ic,'b',t*1e6,1e3*Id,'r--','LineWidth',1.5); grid on;
xlabel('Time (\mus)'); ylabel('Current (mA)');
legend('Conduction current Ic','Displacement current Id');
title('Continuity of current through a capacitor');
```



Test Case (ii) – Conduction and Displacement Current in Different Media

```
clc;
clear;
close all;

eps0 = 8.854e-12;

f = logspace(0,12,500);
w = 2*pi*f;

sigma = [5.8e7  4  1e-3];
er = [1  81  4];

names = {'Copper','Sea water','Dry soil'};

figure;
```

```
for k = 1:3

    ratio = sigma(k)./(w*eps0*er(k));

    loglog(f,ratio,'LineWidth',2);
    hold on;

    fc = sigma(k)/(2*pi*eps0*er(k));

    fprintf('%s: crossover frequency fc = %.4e Hz\n', ...
        names{k},fc);

end

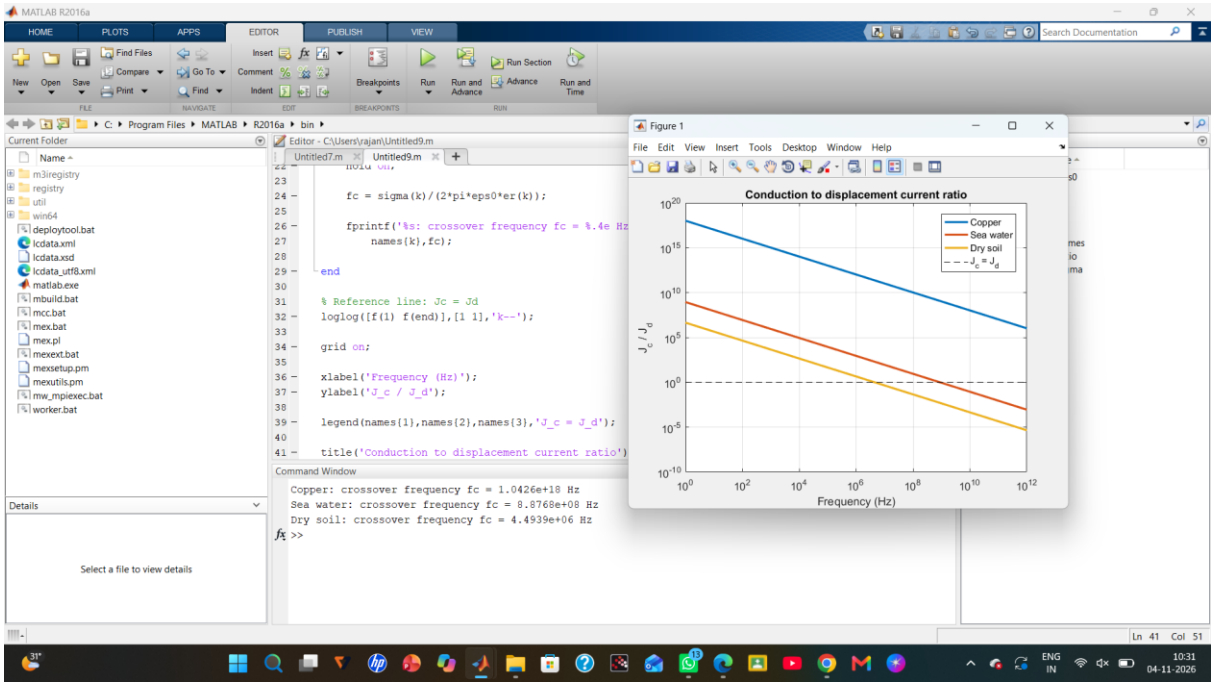
% Reference line: Jc = Jd
loglog([f(1) f(end)],[1 1],'k--');

grid on;

xlabel('Frequency (Hz)');
ylabel('J_c / J_d');

legend(names{1},names{2},names{3},'J_c = J_d');

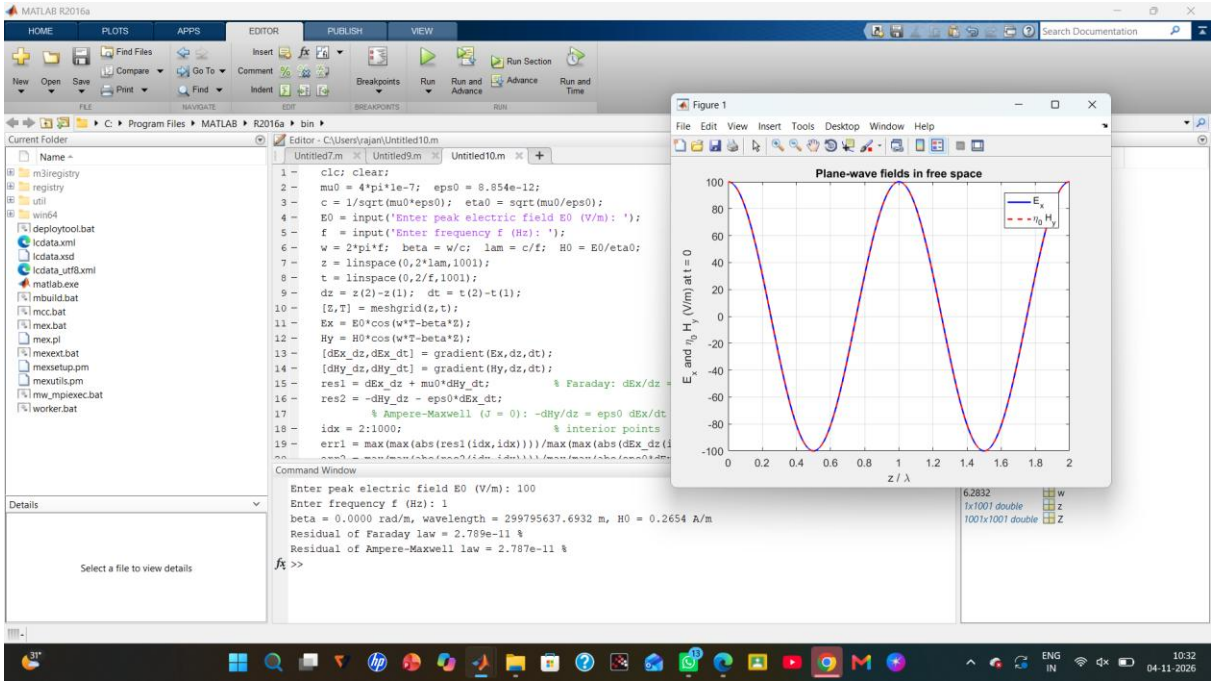
title('Conduction to displacement current ratio');
```



Test Case (iii) – Numerical Verification of Maxwell's Curl Equations for a Plane Wave

```
clc; clear;
mu0 = 4*pi*1e-7;  eps0 = 8.854e-12;
c = 1/sqrt(mu0*eps0);  eta0 = sqrt(mu0/eps0);
E0 = input('Enter peak electric field E0 (V/m): ');
f  = input('Enter frequency f (Hz): ');
w = 2*pi*f;  beta = w/c;  lam = c/f;  H0 = E0/eta0;
z = linspace(0,2*lam,1001);
t = linspace(0,2/f,1001);
dz = z(2)-z(1);  dt = t(2)-t(1);
[Z,T] = meshgrid(z,t);
Ex = E0*cos(w*T-beta*Z);
Hy = H0*cos(w*T-beta*Z);
[dEx_dz,dEx_dt] = gradient(Ex,dz,dt);
[dHy_dz,dHy_dt] = gradient(Hy,dz,dt);
```

```
res1 = dEx_dz + mu0*dHy_dt;          % Faraday: dEx/dz = -mu0 dHy/dt
res2 = -dHy_dz - eps0*dEx_dt;
        % Ampere-Maxwell (J = 0): -dHy/dz = eps0 dEx/dt
idx = 2:1000;                        % interior points
err1 = max(max(abs(res1(idx,idx))))/max(max(abs(dEx_dz(idx,idx))))*100;
err2 = max(max(abs(res2(idx,idx))))/max(max(abs(eps0*dEx_dt(idx,idx))))*100;
fprintf('beta = %.4f rad/m, wavelength = %.4f m, H0 = %.4f A/m\n', beta, lam, H0);
fprintf('Residual of Faraday law = %.3e %%\n', err1);
fprintf('Residual of Ampere-Maxwell law = %.3e %%\n', err2);
plot(z/lam,Ex(1,:),'b',z/lam,eta0*Hy(1,:),'r--','LineWidth',1.5); grid on;
xlabel('z / \lambda'); ylabel('E_x and \eta_0 H_y (V/m) at t = 0');
legend('E_x','\eta_0 H_y');
title('Plane-wave fields in free space');
```



Test Case (iv) – Magnetic Field Produced by Displacement Current (Ampère–Maxwell Law)

```
clc;
```

```
clear;
close all;

eps0 = 8.854e-12;

a = 0.05;
d = 1e-3;
er = 1;
V0 = 100;
f = 1e6;

w = 2*pi*f;
eps = eps0*er;

% Peak displacement current density
Jd_m = eps*w*V0/d;

% Peak displacement current
Id_m = Jd_m*pi*a^2;

% Radial distance
rho = linspace(0.001,0.15,300);

% Initialize magnetic field
H = zeros(size(rho));
```

```
% Inside the capacitor plates
inside = rho <= a;

H(inside) = Jd_m*rho(inside)/2;

% Outside the capacitor plates
H(~inside) = Id_m./(2*pi*rho(~inside));

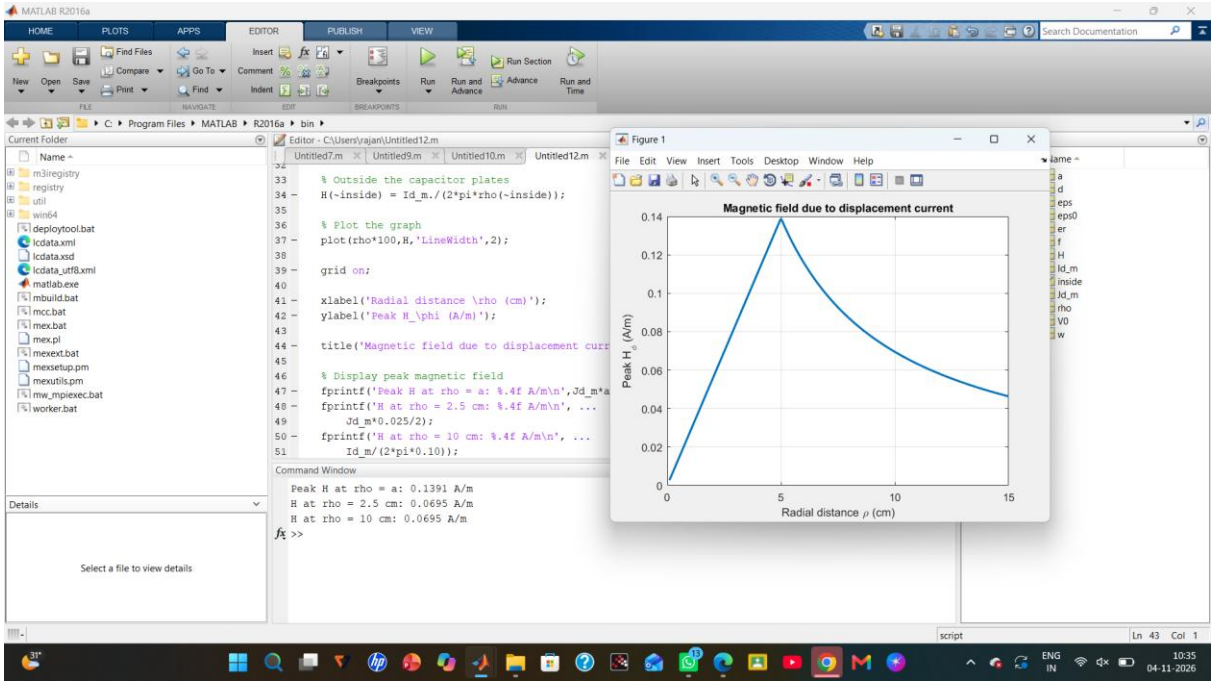
% Plot the graph
plot(rho*100,H,'LineWidth',2);

grid on;

xlabel('Radial distance \rho (cm)');
ylabel('Peak H_\phi (A/m)');

title('Magnetic field due to displacement current');

% Display peak magnetic field
fprintf('Peak H at rho = a: %.4f A/m\n',Jd_m*a/2);
fprintf('H at rho = 2.5 cm: %.4f A/m\n', ...
    Jd_m*0.025/2);
fprintf('H at rho = 10 cm: %.4f A/m\n', ...
    Id_m/(2*pi*0.10));
```



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify Maxwell relations and assumptio ns	Identify Maxwell relations and assumption s	Selects the appropriate integral/differe nt ial equation and identifies time varying quantities.	Selectio n is correct with minor omissio ns.	Needs guidance choosing relations.	Uses inappropriate equations.	25	2.5	
2	Calculate displaceme nt current and field quantities	Calculate displaceme nt current and field quantities	Implements the displacement current relation and associated field calculations correctly.	Calculat ions are mostly correct with minor errors	Partial calculations require guidance.	Calculations are incorrect or absent.	25	2.5	
3	Verify equations numericall y or graphically	Verify equations numericall y or graphically	Demonstrate s consistency of the chosen Maxwell relation with computed values/plots and units.	Verifica tion is mostly correct	Verification is incomplete.	No valid verification is demonstrated .	30	3	

4	Explain physical significanc e	Explain physical significanc e	Explains displacement current and its role in time varying fields with clear documentati on.	Explana tion is generall y correct.	Explanation is superficial or incomplete.	Cannot explain the result.	20	2	
Tota l							100	10	